

CHAPTER 2

Nanoparticles: their applications, limitations and future challenges in medicine

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Abstract: Nanotechnology advancements have led to the development of its allied fields, such as nanoparticle synthesis and their applications in biomedicine. Nanotechnology driven innovations have given hope to the patients as well as physicians in solving complex medical problems. Nanoparticles have extensive applications in the medical field as nanomedicine. That is possible in novel diagnostic instruments, imaging methodology towards the targeted medicinal products, pharmaceutical products, biomedical implants as well as tissue engineering during cosmetic applications. Nowadays treatment of high toxicity could be administered with improved safety measures using nanotechnology applications such as chemotherapeutics, cancer drugs/Cancer cell prediction status. Furthermore, there is an enhanced feasibility on wearable gadgets for abnormal science in the body cell proliferation/ abnormality, sometimes injection which are caused by microorganisms. We anticipate the above-mentioned techniques to provide doctors with considerably better access to obtain critical data for the changes in the life of illness, because of the technological presence at the source of the problem. Biomedicines can also be utilized for therapies with predictive analysis as well as artificial intelligence play an important role.

Key words: Nanoparticles, Treatment, Drug delivery, Cancer therapy.

2.1 Introduction

Nanotechnology is a multidisciplinary unit composed of physics, chemistry engineering and biology to design principles and synthesize the nanoscale devices or materials [1]. The development of nanoscale sized objects and the rapid progress in nanotechnology profusely intend to new hope in promoting human health and veterinary to address several new diseases in future courses of action. Currently, it is recorded as a great concern for the potential health risks. Thus, continuous exposure to nanoparticles develops inflammation, toxicity apoptosis, oxidative stress, and lung infection leading to pulmonary fibrosis, metal fume fever, mesothelioma and ardystil syndrome [2,3].

Usually, the two types of NPs, Nanoengineered found in the environment i.e., normal events like erosion, volcanic eruptions etc., and other engineered ones are the products of physical, chemical, biological and manmade have the strength, performance, and flexibility toward the alternation drown in the field of medicine and other allied sciences [4].

2.2 Metal Nanoparticles

The synthesis of metal nanoparticles is one of the top downs & bottom up entities and different approaches for the synthesis of nanoparticles are depicted in **Figure 2.1**. The first approach has the size reduction by the various physical and chemical treatments. For eg: Imperfections are introduced for top-down approaches. This is one of the major limitations because the surface chemistry and other physical properties of nanoparticles are lightly dependent on surface structure.

In the bottom-up approach, the small entities have joined with each other and fabricated nanoparticles. Where the smaller entities formed are then assembled to produce final particles which are nanometers in size [5-7]. Both biological & chemical approaches lined up for bottom-up approaches. The Nanoparticles synthesized by physical and chemical methods are eco-hazardous, energy-intensive but toxic. Whereas biological methods over the physical& chemical are cost effective eco-friendly, single step processes squired large scale production [8]. No need for energy, pressure, or temperature use of toxicants. Biological methods include the use of microorganisms the NPs have been synthesized which is readily scalable, compatible, eco-

friendly and has a wide range of medical applications. However, the isolation and handling of microorganisms are a little expensive. In this regard, the synthesis of plant mediated metal nanoparticles economic value alternative for the large scale production of metal NPs. For this reason, the plant extracts act as both for the reduction process and also as stabilizers for metal nanoparticles. The plant Neem (*Azadiracktha indica*) is very famous and found in India and other subcontinents. It is one of the important medicinal plants since ancient days. The biological activity of neem is a very wide spectrum; even parts like stem, roots leaves, fruits etc. have medicinal values routinely used for household remedies against several diseases like antipyretic, antitumor anti-inflammatory antibacterial and pesticide activity [9]. The bio-molecules present in neem extracts are terpenoids, nimbaflaroone and sugars etc. These biomolecules are responsible for metal ion reduction with the help of capping agents because these neem plants are available very easily and cost effective [10]. Hence the extracts can be useful for the synthesis of nanoparticles.

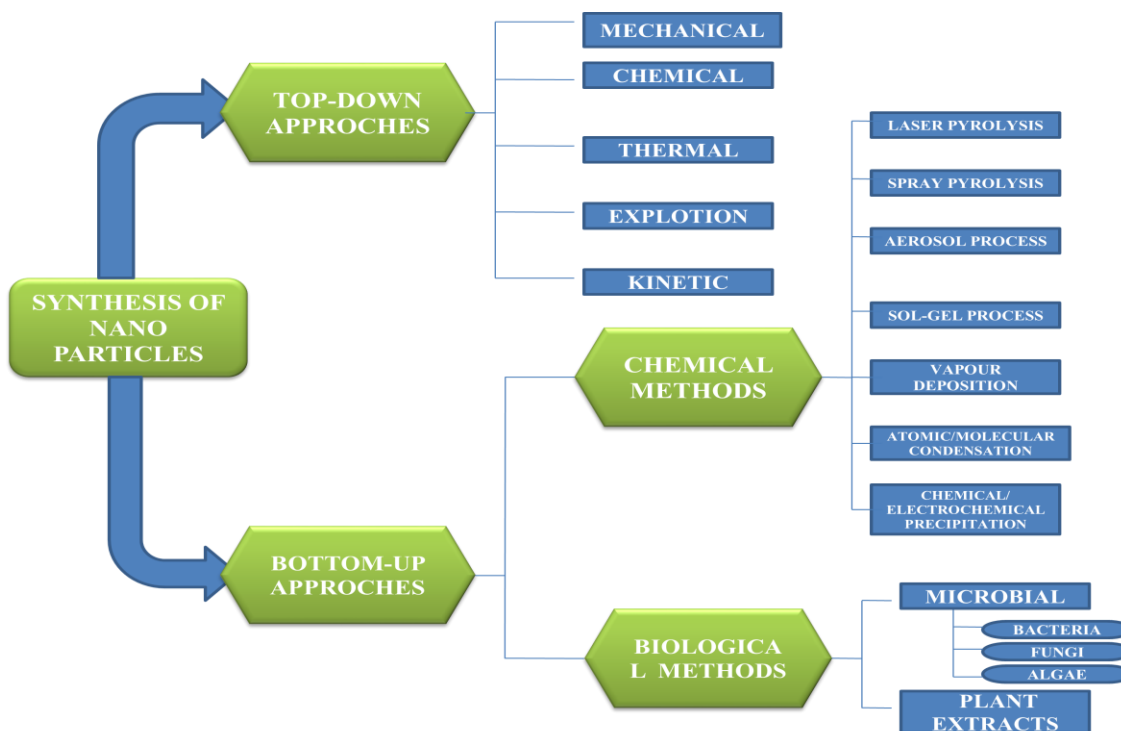


Figure 2.1. Different approaches for the synthesis of metal Nanoparticles.

2.3 Physical Chemical Properties of Metal Nanoparticles

The nanoparticles properties such as size shape surface, morphology, the diameter of the particle and affect the physical stability of NP and the performance of NP *in vivo* [11]. The surface of nanoparticles enhanced with the size reduction and the increased surface to volume ratio, which is helpful to correlate the antimicrobial activity. In the biomedical field, different types of engineered nanoparticles ranging from 0.5 to 1000 nm have been synthesized and characterized for their applicability. Further, the ENp efficacy depends on surface features viz hydrophobicity and surface charge. The nanoparticles interaction with bioactive compounds depends on the surface charge, intensity, and electrostatic value to help in the determination of charge and surface hydrophobicity [12]. Based on the dimensions ENP can be classified as 1D (thin film), 2D (Nanotubes), 3D (dendrimers, quantum dots) [13,14]. Different types of nanoparticles have been developed as well as characterized for the use in drug delivery of bioactive agent system which includes liposome [15], polymeric nanoparticles [16], Nanocapsules [17, 18] polymeric micelles [19] and dendrimers [20,21]. Different methods of synthesis of metal nanoparticles are discussed in the previous section. In a biological system, the effectiveness of engineered nanoparticles mainly correlates with their properties and chemical and biological interactions. The ENP applications in image photography and drug delivery systems require stable and explicit control over NP interaction with cells, that perhaps provided by the surface properties. Moreover, stability and size control issues remain challenging during ENP synthesis [22]. Several physico-chemical properties such as high surface to volume ratio, and mechanically strong, active and chemically reactive nanoparticles exclusively create different suitable applications. The following remarkable properties are given below in **Figure 2.2**.

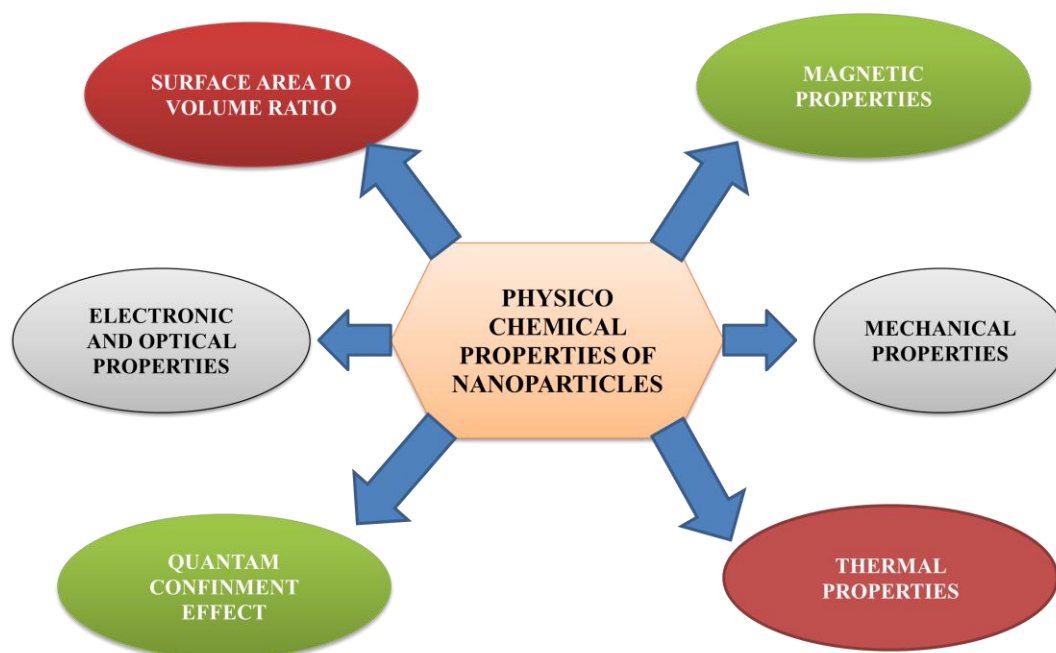


Figure 2.2. Various physical chemical properties of metal nanoparticles.

2.4 Potential applications of nanoparticles in the field of medicine

Nanoparticles and the ENP development have revolutionized in medical field and have given a ray of hope to the physicians in combating the disease still do not have the specific medication. Thus ENP find potential applications in the fields of medicine and biology. This is why several groups have been synthesized and characterized. But few of the inorganic ENP like aluminum, copper gold and magnesium nanoparticles and carbon based nanoparticles has been evaluated for their biological activity. Hence, the development in the genetics, molecular biology, nanotechnology and properties of different DNA based Np's have been a significant impact during the drug delivery, DNA chips and biosensors [23]. Several organic ENP'S including quaternary ammonium compounds, chitosan, polysiloxane and triclosan have also been characterized for their activity. Instead, several other metalloid NP'S such as polymer coated bismuth sulfide cadmium, tellurite has employed in the diagnostic assays [24]. The applications of ENP'S in various fields are represented in **Figure 2.3**.

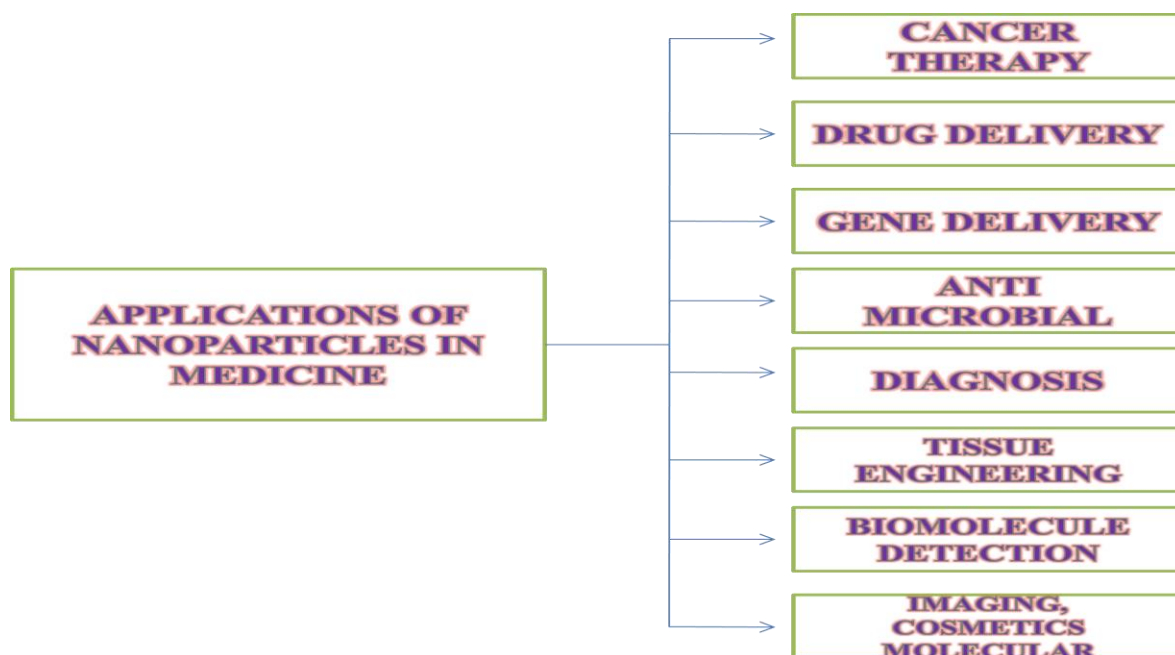


Figure 2.3 Potential applications of nanoparticles in various fields of medicine

2.4.1 Cancer therapy

The traditional chemotherapeutic drugs make them to deliver sufficient quantities because of the tumor site without any risk i.e., with the intrinsic factors limits (water solution, drug resistance, cytotoxicity etc.) Cho et al. 2008 [25]. Drug design and applications for effective therapy for effective therapeutic delivery to the tumor site attract alternation [26]. The microenvironment of the tumor is different from the surrounding normal cells, fast growing cells (Cancer) would show high metabolic rate. The supply of nutrients and O₂ is normally insufficient for growth, so the tumor cells can have glycolysis to obtain more energy which leads to an acidic environment. [27]. Moreover, the microenvironment of the tumor exclusively presents high conc, in general of the physiological reaction and the reduction of (GSH) glutathione agent in particular. As the GSH concentration is found to be high i.e., at least 5 times more than the normal in regular cells, that ultimately costs redox environment in tumor [28].

Therefore, this is high time to design. stimuli responsive drug delivery system, based on the feature specially targeted for tumor cells. However, all these have to be micro-sized it is very much complemented by the rapid development in nanoparticle synthesis and also the surface chemistry [29]. Mono-dispersed magnetic particles with optimal size, composition and surface

can be easily prepared harboring both effective drug loading and stimuli responsive functions. Since all these decades' advances have been promoted towards the application of magnetic nanoparticles as drug carriers. The potential advantages of magnetic NPs as a new paradigm in cancer therapy i.e., 1) Unique properties of their magnetic cores combined with high drug loading capacity. 2) The flexible biochemical properties tend to be obtained utilizing suitable and biocompatible coating. 3) The high range of tumor-specific delivery efficacy minimizes the side effects enormously. 4) The programmed release of the drug via specific stimuli either exogenous (Temperature high and magnetic) or endogenous (pH, Enzyme concentration or redox gradient). 5). The capability of the magnetic probe for reactive monitoring the drug delivery. One of the major problems during drug delivery is the poor site-specific efficacy. To improve and reduce the potential toxicity magnetic controlled targeting strategy is often used. For example, Huhn et al. 2016 [30] reported that the magnetic fluid loaded cytokines could be delivered to the specific organisms by high energy magnetic field i.e., they bound with microxanthane was successfully used in targeting squamous cell carcinoma via magnetically controlled drug delivery. In addition, the effective tumor targeting system drug delivery system requires Zero release before reaching the target site. Hence the stimuli responsive will be highly desirable, especially for the release of higher toxic anti-tumor drugs. Magnetic nanoparticles carrum can realize remote control via pH redox potential higher frequency of magnetic field and induction heat once it was delivered to a specific site. Reported that pH sensitive doxorubicin loaded thermally cross linked super para magnetic 10 oxides nanoparticles. The dox could be incorporated in the polymeric shell of TCL-SPION through electrostatic indolence between positively charged particles and negatively charged ones and approximately 60 % of the drug was released on or before one hour at minimum acidic in acetate buffer ($\text{pH} \leq 5.1$). In acetate buffer Sun et al. 2006 [31] reported that the pH responsive porous hollow nanoparticles application for breast cancer, exploited as delivery of cisplatin through a vehicle targeted for therapeutic use by BR-3 cells. The acidic etching of porous hallows nanoparticles results the wider pore gaps which facilitates the release of cisplatin up to 4 times greater than the normal physiological conditions i.e., ph. Lin et.al., 2015 [32] published a report on redox responsive system in the delivery systems which is composed of disulfide linkages of MCM-41- type mesoporous silica nanorods (MSNR) and the Fe_3O_4 nanoparticles. The disulfide linkages can be cleared by various cell produced antioxidants as well as disulfide agents with reducing probability, which results in the release of the magnetic

nanoparticle caps from the MSNS that trigger the escape of the molecules in the matrix. The magnetic induction and genetic field used stimuli in response to drug release. Chen et al. 2013 [33] reported magnetic silica based Nano shapes as effective drug carriers for the controlled release of molecules by magnetic heat. This HFMF accelerates the magnetic rotation where nanoparticles that induce heat energy subsequently extend the structure of the silica matrix producing a porous structure that releases the drug very easily Hu et.al., 2008 [34]. They used the ferrous oxide core coated with sensitive gel with the temperature that encapsulates the anticancer agent. As and when the inductive heat collapses the nanoparticles in oxide form, resulting in an accelerative drug dispense. Liu et al. 2008 [35] reported among the magnetic nanoparticles that induce heat responsive fluorophore amine release, the super and magnetic material of iron single crystalline capsule.

2.4.2 Drug delivery

Researchers found in 2001 that Mesoporous Silica Nanoparticles (MSNs) have enormous potential as a drug delivery method [36]. Single nucleotide polymorphisms (SNPs) have since been widely used in the administration of drugs [37]. There are several SNP-based drug delivery methods, including quick, sustained, stimulus-response control, and customized drug delivery systems (**Table 2.1**). Additional research successfully manufactured a controlled release system comprised of MSNs with covalently bonded dipalmitoyl moieties that support phosphorylated lipids. The pores of lipid bilayer-coated MSNs (LB-MSNs) can emit fluorescein by chemically induced disulfide reduction. The use of LB-MSNs as a regulated medication delivery method has significant promise. In 2013, researchers created a medication delivery method for cancer that employs photosensitive, cooperative therapy with cytotoxic cadmium sulfide nanoparticles and camptothecin [38]. Zhao et al. [39] optimized the system utilizing Cy-5.5-labeled silica cross-linked PEO132-PPO50-PEO132 micelles with a dynamic particle size of 14 nm.

These micelles provide a stronger drug diffusion barrier from the polymer core to the silica cross-linking layer, with a half-life of up to 19 hours in blood circulation. As shown in **Figure 2.3**, this leads to efficient accumulation via the well-known enhanced permeability and retention effect in tumor models implanted subcutaneously [40]. Thyroid-stimulating hormone receptor-containing SNPs were proposed by Li, Zhang et al. [41] as potential targets for thyroid cancer.

The researchers have successfully designed nanoparticles that can cure cancer using a selective etching technique based on structural variations sensitive to glutathione (GSH). The hollow mesoporous NPs (HMSiO₂ NPs) can release a drug payload from doxorubicin (DOX) triggered by acid. The study demonstrated that HMSiO₂ NPs have a high load capacity and controlled breakdown at a concentration of 100 µg/ml. They can efficiently kill cancer cells by containing 9 µg/ml of DOX. The production methods, DOX release curves, characterization, and in vitro cytotoxicity of HMSiO₂ NPs were compared to those of non-GSH-sensitive TEOS HMSiO₂. Results indicate that HMSiO₂ NPs show great promise for cancer treatment. Cornell prime dots, also known as ultra-small fluorescent core-shell SNPs, were functionalized by Juthani et al. [42] using PET labels (124I, 89Zr) and either integrin-binding (cRGDY) or nontargeting (cRADY) peptides. Next, using a genetically altered mouse model of glioblastoma (GB), they investigated the possibility of using dual-modality cRGD-C0 dots to enhance collection, distribution, and retention (ADR). This enhanced the transport and penetration of drugs for brain tumors as well as ADR. Cerium oxide nanoparticle-coated MSNs (COP@MSNs) were developed by Singh et al. [43] and employed a pH-triggered response mechanism to control drug release and intracellular drug delivery. The carboxyl-functionalized MSNs' mesopores were blocked with aminated COP. The pores opened in an acidic environment, releasing the loaded medicines and forming a pH-responsive drug delivery system. Due to their unique characteristics and reaction mechanisms (pH, light, magnetic, etc.), SNPs have the potential to be helpful in medicine, especially in the treatment of tumors.

2.4.3 Gene delivery

Metal nanoparticles are a highly effective platform for gene delivery. This cutting-edge technology harnesses the unique properties of gold, silver, platinum, palladium, and other nanoparticles to deliver genes with unparalleled efficiency and precision. Their distinctive optical absorption and scattering properties, arising from localized Surface Plasmon Resonance (SPR), render them innovative for use in bio imaging or biomedical contexts. Because of their easily synthesized nature, versatility in surface modification, bio-conjugation, adjustable optical properties, and capacity to change size and form, gold and silver nanoparticles (NPs) are very interesting. Metal nanoparticles are a strong technique for transporting a range of substances,

Table 2.1. Composition of the SNPs based drug delivery systems.

Drug delivery system	Type	Characteristic	Model drug
Immediate drug delivery system (IDDSS)	--	Improving dissolution rate of hydrophobic drugs	Telmisartan (TEL) Prednisolone
Sustained drug delivery system (SDDSs)	Unmodified modified	Providing long-term drug release	Felodipine, nonsteroidal anti-inflammatory etc
Stimuli-responsive controlled drug delivery system (CDDSs)	Endogenous: pH redox, enzyme, glucose, H ₂ O ₂ , ATP Exogenous: Thermo, light magnetic ultrasound, electro	Effective control of drug release	β -Cyclodextrin, dexamethasone.
Targeted drug delivery system (TDDSs)		Targeted delivery of drugs, less dosage, tumor drug	Doxorubicin, camptothecin, TNF- α protein etc.

including DNA, drugs, and peptides, directly into animal/mammalian cells. Their potential to transfer biomolecules into plant cells is mainly unrealized, despite their success as micronutrients or Nano-fertilizers to increase agricultural yields. The primary use for metal nanoparticles has been as triggers for cargo release and as capping agents for other nanomaterials. To show, Vijayakumar and his associates used carbon bullets loaded with gold nanoparticles to actively transfer genes. It was made feasible by a genetic gun. Gold nanoparticles were produced after incubation with fungal biomass. Ruby-red coloration from the reduction of gold ions showed that the synthesis of AuNPs had been effective. At high temperatures ($\sim 600^\circ\text{C}$), desiccating biomass with Au-NPs improves crystallinity. Furthermore, it has been shown that these particles are encased in a carbon matrix and may transport up to 2400 ng of plasmid DNA. The addition of gold nanoparticles in a composite increased the available surface area for cargo loading, provided high packing density, and reduced nuclease degradation. By introducing HTC 600-Au plasmid DNA carrying a GUS reporter expression cassette, rice, and tobacco cells were altered. The GUS reporter enzyme's expression was measured to assess the transformation.

DNA delivery into the plant cell was made possible in large part by the composite material's sharp-edged carbon matrix. Without gold, the pure-carbon substance proved inefficient at attaching the DNA. The overall conclusion of this study was that carbon bullets coated with nanogold can reduce cell damage and increase the effectiveness of plant cell transformation [86]. To transfer DNA into plant cells, Yu-qin et al. [44] employed ZnS nanoparticles, which had a size range of around 3-5 nm. Positively charged Poly-L-lysine (PLL) was an efficient way to functionalize zinc sulfide nanoparticles (ZnS NPs) such that they could connect to DNA through electrostatic interactions. The efficient use of ultrasound enabled the transport of plasmid-coated ZnS NPs into tobacco cells, culminating in the successful development of stable transgenic tobacco plants. Furthermore, the scientists proposed that PLL-coated gold nanoparticles might protect DNA from ultrasonic damage. Because of their low loading capacity, metal nanoparticles are seldom employed for gene delivery in plants. However, they are extensively used for bio-imaging and cargo release (**Table 2.3**). Magnetic Nanoparticles (MNPs) have magnetic characteristics in addition to their normal features as metal NPs. MNPs consist of metal oxides (iron oxides), metal alloys (FeCo, FePt, etc.), and metal nanoparticles (nickel or cobalt). MNPs are superparamagnetic alloy and exceptionally biocompatible. "They only exhibit magnetic characteristics in the presence of a magnetic field." MNPs with positively charged surfaces have shown great potential as a non-assisted nucleic acid delivery vehicle in both plant and animal cells. Higher digestion resistance, greater nucleic acid loading capacity, enhanced penetration, significant cost savings, and the ability to transfer biomolecules when exposed to an external magnetic field are just a few advantages of superparamagnetic nanoparticles.

Using magnetic force to move MNPs loaded with cargo into cells, magnetofection is an efficient biomolecule delivery technique. It doesn't require complex instrumentation and is simple to implement. Hao et al. [45] described the use of a magnetic gold nanoparticle (mGNP) compound as a plant cell gene transporter. PEG was utilized in a surface functionalization procedure to produce mGNPs with a diameter of about 20–30 nm. Plasmid DNA and fluorescein Isothiocyanate (FITC) were covalently attached to the surface of mGNPs via an amide bond, and the molecules were introduced into canola protoplasts using an external magnetic field. It has been observed that FITC-labelled mGNPs can deliver up to 95% of their target in canola

protoplasts. Similarly, Zhao et al. [46] reported on an MNP-based strategy for plant transformation known as "pollen magnetofection" DNA delivery. The model plant cotton's pollen grains have huge holes and are protected by extraordinarily porous thin walls, which allow foreign proteins to enter. Using polyethyleneimine (PEI) to functionalize the magnetic nanoparticles, cationic magnetic nanoparticles were loaded onto exogenous DNA to form MNP-DNA complexes. Healthy cotton pollen was mixed with MNP-DNA complexes and then moved via an external magnetic field. Following plant fertilization using magnetofected pollen, transgenic seeds were created without the need for tissue culture. Notably, transgenic plants showed that foreign DNA was gradually handed down to offspring and that the transgene was both incorporated into the plant genome and expressed. Magnetofection is a straightforward process that fully removes the need for tissue culture. It can modify several genes quickly.

But since magnetoreception efficacy is directly impacted by the quantity and size of pollen apertures, it will be challenging to modify pollen with a single, tiny hole using this method. However, the method developed by Zhao and associates raises the intriguing prospect of inserting foreign DNA into the plant genome and creating transgenic plants without requiring the tissue culture phase of plant genetic engineering. Superparamagnetic Fe₃O₄ nanoparticles' surfaces are normally modified by applying positively charged polymers, including polyethyleneimine (PEI). It prevents cargo degradation by facilitating negatively charged DNA binding to nanoparticles through electrostatic interactions.

Its ability to react to an external magnetic pull is what makes it superparamagnetic; this property helps with cargo internalization and enhances transformation efficiency. Numerous studies have demonstrated the viability of transferring genes and metabolites in plants utilizing various nanoparticles, including starch NPs, polymer composites, and quantum dots. To carry genetic material for the delivery of dsRNA in plants, Mitter and colleagues have developed a novel class of inorganic metal ions called layered double hydroxide (LDH) clay Nano sheets that are biodegradable. Complexes of dsRNA-cationic LDH (Bio Clay) were formed by electrostatic loading of dsRNA onto its surface. When the substance was applied to the leaf surfaces of Arabidopsis plants, it entered plant cells and used RNA interference-mediated systemic immunity to shield the plants against viral infection. Carbonic acid formation on the leaf surface

as a result of atmospheric CO₂ and humidity assisted in Bio Clay decomposition and triggered long-term dsRNA release. Apart from its transport function, biodegradable LDH provided a prolonged barrier to protect the dsRNA from nuclease action. Few uses in plant science have been found for the use of RNAi-carrying nanoparticles, which have mostly been investigated for human medications. In plant systems, however, nanomaterial-based RNAi delivery has been made possible by the work of Mitter and colleagues. In contrast, Zhang et al. created novel DNA nanostructures that allow genetic material to be transferred into plant cells without the need for outside mechanical assistance.

Tests on a range of plant species, including *Nicotiana benthamiana* (Nb), *Nicotiana tabacum*, and *Eruca*, were successful using these nanostructures. This study shows that a variety of nanostructures containing genetic material can be internalized by plant cells. The size, shape, and compactness of nanostructures may directly affect the internalization of nanoparticles (NPs) in plant cells, even if the mechanism of NP internalization is unknown. Fluorescence-labeled starch nanoparticles were employed by Liu et al. [47] to introduce DNA into plant cells. For five minutes, suspension cells of *Dioscorea* spp. were co-cultured with DNA-loaded starch nanoparticles and exposed to ultrasonic vibrations (120W at 40 kHz). To track the dispersion of nanoparticles in plant cells, a fluorescent tag was employed. The size of the DNA-starch NP complex varied between 50 and 100 nm, and it is hypothesized that it entered the cell through a channel formed by ultrasonic vibrations. Starch NPs interact with DNA to protect it against DNase I cleavage as well as ultrasonic cleavage. To transport DNA to plants, starch-NPs are not frequently employed because of their poor transformation efficiency. The transformation effectiveness of *Agrobacterium tumefaciens* is around 54%, whereas the gene delivery method utilizing calcium phosphate nanoparticles has been reported to have an efficiency of over 80%.

Wang et al. [48] examined the visualization of nanoparticle-mediated gene transfer in plants using fluorescent quantum dots (CdSe). The author effectively transported genes into the *Jatropha curcas* callus using chitosan-DNA nanoparticles. To identify nanoparticle-mediated gene transfer, CdSe quantum dots were "labeled over chitosan-DNA nanoparticles complex". The use of silicon carbide (SiC)-mediated DNA delivery is one of these nanoparticle methods that may hasten the transfer of nucleic acids to plant cells, opening the door for non-conventional

materials to deliver biomolecules to plant cells^{46, 100}. SiC whiskers have various benefits, such as higher tensile strength, higher elastic modulus, higher intrinsic hardness, and superior resilience to shock and degradation, which have led to extensive research into a wide range of applications. A SiC-based genetic transformation approach for plants was presented by Wang et al. [48]. Recently, SiC whiskers were used to genetically modify peanuts to introduce the chitinase gene. The transgenic plants that emerged were resistant to leaf spot disease.

2.4.4 Antimicrobial

Despite several investigations over the years, the particular mechanism of action for AgNPs is unclear. AgNPs have antibacterial effects through four main mechanisms: (i) adhesion to the surface of bacteria; (ii) disruption of the bacterial cell wall and membrane through permeability changes; (iii) oxidative stress induction through the production of ROS and free radicals; and (iv) modification of signal transduction pathways [49]. Many writers describe AgNP adherence to bacterial surfaces as the first step of a complex bacterial inhibitory process. AgNPs' adherence is significantly influenced by their size and zeta potential. AgNPs can be positively, neutrally, or negatively charged on the surface, depending on how they were made. The findings of Abbaszadegan et al. showed that altering the surface. The slightly negative charge on the surface of the bacterium attracts positively charged AgNPs, which enhances the antibacterial activity of the bacteria (**Table 2.2**). Negatively or neutrally charged nanoparticles, however, have far less of an antibacterial impact. The decrease of electrostatic repulsion can be achieved by bacterial surface saturation technique, though, if AgNP concentration is increased [50]. AgNPs can engage in two different types of cell interaction after adhering to the bacterial surface. Whereas bigger nanoparticles stay outside the bacteria, smaller AgNPs get straight inside the cell. In both circumstances, AgNPs constantly release Ag⁺ ions. Proton leakage results from the membrane potential being unstable when these ions adhere to the architecture of cell membranes. Cell wall instability significantly improves bacterial permeability, enabling bigger AgNPs into the cell [51]. Cell dysfunction is brought on by interactions between AgNPs and Ag⁺ ions and a range of biomolecules and structures found within the cell, such as proteins, lipids, and DNA. AgNPs are well known for their capacity to produce free radicals and reactive oxygen species (ROS), such as hydroxyl radical (OH•), superoxide anion (O₂⁻), and hydrogen peroxide (H₂O₂). Even though ROS are a normal byproduct of cellular respiration in bacteria, these organisms possess

inherent defensive mechanisms that function as antioxidant enzymes and eliminate pathogenic species. AgNPs produce large levels of Ag^+ , which significantly increases the stress response. Some of the released ions are removed by antioxidant enzymes, but not enough is done so to balance the silver nitrate concentration [52]. These species interact with membrane-bound respiratory chain proteins, inactivating enzymes, due to their great attraction for phosphate, thiol, and carboxyl groups [53]. Their bonding with phosphate groups frequently contributes to enzymatic activation and inhibits the phosphorylation of proteins, which in turn slows down the growth of bacteria. The disruption of the cell cycle has also been linked to the dephosphorylating of tyrosine residues in proteins, which can stop the formation of exopolysaccharides and capsular polysaccharides and their transit to the membrane [62]. Ag^+ can also intercalate DNA strands by forming complexes with nucleic acids and rupturing the H-bonds that hold purine and pyrimidine base pairs together [52].

2.4.5 Diagnosis

Identification of infectious diseases by interacting with substances that bind to nucleic acids preferentially and are readily separated by magnetic fields, MBs provide a special technical approach for nucleic acid extraction.

Table 2.2. Summary of infectious detection

Magnetic particles	Experimental sample	Detection limit
MBs	Group B streptococci	1.25×10^3 cfu/ml
GMBN	<i>S. cholerae</i>	5×10^5 cfu/ml
AuMagNBs	Influenza virus A	4.42×10^{-14} g/ml
MNPs	H9N2, H1N1, H7N9	2×10^{-2} pg/ml

Parham et al. [54] devised a sensitive and effective DNA capture approach that may be used in conjunction with PCR and non-amplification detection techniques. To gather GBS genomic DNA (gDNA), functionalized superparamagnetic beads were exposed to varying lengths of oligonucleotide capture probes. Quantitative PCR was then used to identify the isolated DNA. Magnetic microbeads with sequence-specific hybridization capture could offer a sensitive and

effective way to identify nucleic acids in a variety of organic materials. This could allow for microfluidic system point-of-care testing.

Zhuang [55] extracted DNA and RNA using a magnetic viral DNA kit. After MB-PCR purification using a purification kit, a fluorescence lateral flow immunoassay was conducted on the PCR products. Wu [56] used fluorescent MNPs to detect H9N2, H1N1, and H7N9 avian influenza viruses simultaneously. MNPs can separate and capture target chains, and fluorescence can be used to count individual particles. This method has excellent sensitivity for early illness diagnosis and is based on straightforward signal amplification techniques. Ali [57] reports that silica-modified MNPs were used to extract and amplify HIV, HBV, and HCV viral nucleic acids. By cultivating MNPs modified with three distinct probes in separate tubes and exposing them to RT-PCR biotin-labeled products, the HBV, HCV, and HIV sequences were obtained. To measure the chemiluminescence (CL), the final step included adding 3-(2'-spiro adamantane)-4-methoxy-4-(3'-phosphoryloxy) phenyl-1,2-dioxetane. This approach allows for the simultaneous, high-throughput, automated detection of several nucleotide sequences. Using electrochemical magnetic microbeads, María created a biosensor [58]. The platform is inexpensive, sturdy, and lightweight. Studies have indicated that it can be employed for the detection and diagnosis of illnesses in humans (such as brucellosis and Chagas disease) as well as illnesses in animals. It is possible to diagnose a wide range of infectious and non-infectious diseases using this platform's technologies. The immunochromatographic test strip can be used with food samples, such as whole milk. To diagnose infections, Jacquelyn [59] suggested employing microfluidic manufacturing technology in conjunction with MB aggregation. In this investigation, the LAMP product was employed in concert with product inhibition bead aggregation to determine the target sequence of *Clostridium difficile*.

Chen [60] employed MNPs to develop a one-of-a-kind point-of-care testing (POCT) system. After the performance of the equipment was assessed. The testing findings, which are comparable to those of commercial systems, show how reliable the system is. Margarita [61] extracted DNA and Leishmania-specific surface antigens from the mixture using MB capture probes before determining the presence of cadmium selenite quantum dots. These techniques have a lot of potential for use in human and veterinary medicine. Gold nanoparticles and

magnetic Nano beads in silica shells were recently used by Sangjin [62] to perform an enzyme-linked immunosorbent test to track the influenza A virus, halt its spread, and offer prompt therapeutic therapy. *Yersinia pestis* was detected in liver samples treated with MBs from Feng [63] using LAMP. When combined, MBs and LAMP offer a simple, efficient, and precise detection technique that works well in fields or underprivileged areas. With the help of a warming and vibration module, Fang [64] created a high-flux magnetic separation module to collect a large number of purified nucleic acids from pathogens of infectious diseases. To effectively treat Alzheimer's disease, it is essential to identify and quantify soluble AD biomarkers as early as possible. Examples of these biomarkers are total tau protein and A β concentrations in CSF or plasma. However, as noted by Brambilla et al. (2011) [65], these two methods may yield imprecise and confusing results because of low concentrations and other factors. Yang et al. (2011a) [66] have documented the use of magnetic reagents linked with antibodies against A β -40 and A β -42 to identify A β in blood. According to Yang et al. (2011a), [66] the immunomagnetic reduction (IMR) test has high specificity and can detect concentrations as low as 10 pg/ml at A β -40 and 20 pg/ml at A β -42. According to Skaat et al. (2013) [67] aA β mAb clone BAM10 may be fixed to near-infrared fluorescent maghemite nanoparticles, which facilitates the ex vivo fluorescence and magnetic resonance imaging (MRI) detection of A β 40 fibrils.

Zhou et al. [68] developed DDNP-SPIONs, which bind strongly to A β (1-40) aggregates in vitro using fluorophotometry. DDNP-SPIONs have the potential to be utilized in the molecular diagnosis of Alzheimer's disease due to their compact size, monodisperse size distribution, and changeable surface chemistry. In their work, Jiang et al. [69] utilized BaYF5: YbEr nanoparticles (UCNPs) as up conversion fluorescent tags and Fe₃O₄ MNPs as recognition and concentration components in combination with the A β oligomer aptamer and its corresponding oligonucleotide. An effective approach for quantifying A β oligomers in synthetic CSF was developed by Jiang and colleagues (2017)[70]. PEG-b-AGE, an anti-biofouling polymer, coated magnetic IONPs efficiently extract tau protein and A β 40 and A β 42 peptides from CSF and blood samples with high specificity (>90%) and sensitivity (>95%)

2.4.6. Parkinson Disease

According to Yang and colleagues [71] α -synuclein may be detected in plasma at concentrations ranging from 0.3 fg/ml to 310 pg/ml utilizing an ultrasensitive immunoassay that uses IMR. Based on plasma α -synuclein levels, nanoparticles may differentiate between PD, PDD, and healthy individuals. Research has linked multimodal IONPs to Rhodamine-B (MION-Rh) by conjugating them to the blood of umbilical cord MSCs (UC-MSC; Sibov et al., 2014 [72]. After adult male PD rats were given tagged cells to inject into their striatum, the cells migrated through the area known as the substantia nigra to the medial forebrain bundle as less intense patches, as seen by T2 MRI after 15 days Sibov et al. 2014 [72].

Table 2.3. Different Bio-imaging technologies

Bioimaging	Imaging medium	Contrast medium	References
Optical imaging	Fluorescence	Fluorescent dye-conjugated MSNs	Heidegger et al. (2016)
Computed tomography	X-ray	Au@MSNs	Song et al. (2015)
Magnetic resonance imaging	Radio waves	Fe ₃ O ₄ @MSNs	Kim et al. (2008)
Ultrasound imaging	Ultrasonic waves	ELS	Wang et al. (2012)

2.4.7. Cosmetic Applications

Nanoparticles are acknowledged as useful additives to industrially relevant compounds. Effectively used nanoparticles have a variety of special multifunctional qualities, including a high surface/volume ratio, biological activity, stability, overall product efficiency, and catalytic performance. The features of nanoparticles listed above make them especially well-suited to enhancing the product of interest's overall performance [73]. Nanoparticles with varying morphologies and chemical compositions are commonly used in cosmetic formulations to capitalize on the benefits of shape- and size-related activities [74], which can improve product performance. Because of their ability to scatter and reflect UV light, a variety of inorganic oxide-based nanoparticles, including ZnO, TiO₂, ZrO₂, and CeO₂ nanoparticles, are used as sunburn preparations as physical UV filters as opposed to UV absorbers. The UV radiation attenuation characteristics of these nanostructured materials increase their tolerance to UV light [75]. While CeO₂ nanoparticles do not generate reactive free radicals when exposed to UV radiation, they do

exhibit similar UV-blocking qualities to TiO₂ [76]. Shorter UV-A radiations were shown to be more effectively screened by calcium-doped CeO₂ [77]. Moreover, it has been demonstrated that cerium phosphates and cerium-titanium pyrophosphates function effectively in sunscreen formulation as substitutes for ZnO and TiO₂ nanoparticles [78]. Nanoparticles such as clay and silica are used in cosmetic compositions as thickening agents [79]. Furthermore, hybrid nanoscale inorganic-organic compounds are used to make thickening agents in cosmetics. An illustration of this is the application of perfluorocarbon hydrophobically altered alkali-soluble emulsion (HASE) polymer-grafted SiO₂ nanoparticles to form a nanoscale cross-linked polymeric network structure that protects the skin against exposure to and toxicity from organophosphates [80]. Gold nanoparticles, sometimes referred to as nanogold, range in size from 5 to 400 nm. They come in a wide variety of sizes and forms. The resonance frequency of gold nanoparticles is highly dependent on their size, shape, dielectric characteristics, and other environmental factors. The hue of nanogold changed from red to purple, blue, and nearly black due to aggregation [81]. The production of green gold nanoparticles was recently demonstrated by Haddada et al. [82], who evaluated the method's antioxidant effects. The primary characteristics of gold nanoparticles are their great stability, biocompatibility, inertness, and lack of cytotoxicity. Gold nanoparticles have become popular due to their potent antibacterial and antifungal properties. Anti-aging products, face packs, lotions, creams, and deodorants are just a few of the cosmetic products that include these nanoparticles. Antibacterial and anti-inflammatory properties, blood flow acceleration, delaying the aging process, increased skin suppleness and firmness, and skin metabolism revitalization are only a few of the fundamental properties of nanogold found in beauty care products [83]. In addition to emulsions, liposomes, and polymeric nanoparticles, solid lipid nanoparticles (SLN) and nanostructured lipid carriers (NLC) are intriguing carrier systems. Both SLN and NLC have several advantages for topical cosmetic and pharmaceutical applications, including targeted drug administration, controlled release of actives, occlusion, greater drug penetration, and higher skin moisture. Lipid nanoparticles are a "nanoscale" carrier system with considerable tolerability and low cytotoxicity since they are composed of physiological and biodegradable lipids [84]. Furthermore, substances that are susceptible to oxidation, hydrolysis, and light can have their chemical stability increased by lipid nanoparticles. By integrating into the lipid Nano carrier system, increased chemical stability has been achieved for several cosmetic compounds, such as coenzyme Q10, vitamin E

(tocopherol), ascorbyl palmitate, and vitamin A (retinol) [85]. Although there are many uses for nanoparticles, including in nanomedicine and cosmetics, it is necessary to comprehend the mechanisms underlying their interactions with biological moieties to optimize and control their use [86]. Realizing the potential prognostic and therapeutic effects of nanoscale particles requires first ensuring their safe access into cells. As these nanostructured material carriers are intended to transport target molecules like as drugs, genes, or other specific intracellular locations, nanoparticles' intracellular fate is particularly critical. However, there is still a significant challenge in utilizing nanoparticles to achieve controlled and efficient cellular penetration. Beyond their interactions with cell membranes, a comprehensive knowledge of the mechanisms behind the absorption and trafficking of nanoparticles is essential for carefully optimizing their physicochemical properties, which in turn leads to the development of safe and efficacious nanotherapies [86]. At the nano-bio interface, it has been shown that nanoparticles can change a cell's fate, start a conversation between cells, cause or stop mutations, and change the morphology of cells. Neuronal cell lines' intracellular calcium levels can also be impacted by differently designed nanoparticles. For example, at non-toxic doses, SiO₂ nanoparticles have changed the electrical activity of neuroendocrine cells without altering DNA [87].

2.5 Limitations

- Their long term toxicity is not well understood. According to studies, certain nanoparticles may accumulate in the body and damage organs and tissues. Reactive oxygen species produced by NPs are a primary contributor to toxicity and inflammation.
- Drug compounds frequently undergo significant alterations in their bio distribution when they are administered using nanoparticles. Local overexposure may result from the subsequent absorption in certain organs.
- Nanoparticles are a very effective method of delivering medication precisely to the brain, as they might potentially pass through the blood-brain barrier, but there's also a major drawback as drug-carrying nanoparticles could be hazardous to the brain.
- the large surface area of NPs promotes particle agglomeration, followed by formulation instability owing to high surface Gibbs energy, which is extremely difficult to avoid
- It is difficult to independently manage the intrinsic properties of nanoparticles (NPs), which might alter their stability and cause problems such as sedimentation.

2.6 Future challenges

- When it comes to basic concerns like environmental transformation and interactions with other environmental contaminants, the fate and impacts of Nano medical products are little understood. improvements in medication loading and release technology as well as the ongoing exploration of metallic nanoparticles' potential for medical diagnosis and treatment.
- Immunological reactions, which are hard to predict based on studies in small laboratory animals, such as hypersensitivity reactions, which are only observed in a small percentage of humans upon Nano medicine administration but can be quite severe and occasionally even life-threatening, are among the safety issues associated with Nano medicines. Immunological side effects are mostly caused by activation of the complement cascade, and some interactions between blood cells and nanoparticles have also been shown to play a role in this. In vitro, complement binding and cell interaction experiments can be used to resolve these problems, and preclinical safety studies in larger animals are also advised.

2.7 References

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